

Listening quick-start (sdroxide — shortwave & broadcast)

A short, task-oriented walkthrough for using **sdroxide** as a **listener**: browse the broadcast schedule, tune longwave, medium wave and shortwave, log what you hear and send a reception report. Nothing here needs a licence or a transmitter — the listener's screen is a clean receiver by design. For the reference detail behind each control see [USER _MANUAL.md](#); for the why of this fork see the [README](#).

This is the CB/SWL fork of sdroxide. The listening tools are added on top of the amateur program; the amateur side, and everything else, is upstream and unchanged.

Nederlands: [listening-quickstart.nl.md](#). Français: [listening-quickstart.fr.md](#). Italiano: [listening-quickstart.it.md](#). PDF: [listening-quickstart.en.pdf](#).

Bold names such as **SETTINGS** and **SWL LOG** are buttons and fields exactly as they appear on screen. Settings > Radio is a menu path.

What you need

- **An SDR or a radio sdroxide supports.** Any of them will do; for listening the usual choices are:
 - **RTL-SDR** (dongle) — native, no SoapySDR needed.
 - **RX-888 / RX-888 Mk2** — native; the firmware is uploaded automatically.
 - **Airspy HF+** (Dual / Discovery / Ranger) — native, 0.5 kHz–31 MHz.
 - A **receive-only receiver** such as the **ATS Mini** (an ESP32 + Si4732) — the computer tunes it and demodulates nothing itself; the audio comes back through a sound card.
 - Also possible: HackRF, Airspy R2/Mini, SDRplay RSP, ELAD, PlutoSDR, or a **CAT rig** over a serial port + sound card, TCI, OpenHPSDR, or SoapySDR.
- **An antenna.** For shortwave even a long wire works wonders; a broadcast band wants more wire than an 11 m whip.
- **sdroxide installed** (below).

Installing

- **Windows** — the installer (.msi) or the portable .zip: see the [Releases page](#).
- **Linux** — the **AppImage** (one file, chmod +x and run), the .deb, or the portable tarball.
- **macOS** — the .dmg.

You can also run sdioxide as a **server** and open it in a browser: `sdioxide --server`, then `http://localhost:4950`.

Part A — One-time setup

Everything here is saved under `~/.config/sdioxide-brown/`, so you do it once.

1. Pick your radio

Open **SETTINGS** and go to the **Radio** tab. Pick your interface, the **sample rate** and the **gain**. Changes apply right after **Apply / reconnect**. An **ATS Mini** is chosen here too, as a receive-only source.

2. Turn on the listener's screen

Settings > Radio > Transmit controls > SWL mode hides every transmit control — PTT, TUNE, CALL CQ and the rest — and swaps the ham chips for the listener's (**SCHEDULE, SWL LOG**). It is set **per radio**, so a listening set and a transceiver can sit side by side. **Settings > UI > Start in SWL mode**, or `--swl` on the command line, makes every radio a listener. A receive-only radio (a public SDR, an RTL-SDR) is offered **Listening controls** in its warning banner, which does the same.

3. Fill in your details

Go to the **General** tab:

- **Locator / grid** — your Maidenhead grid. The SWL LOG window pre-fills each reception's **locator** from it.
- **IARU region** — the region that sets the band plan.
- On the **Spots** tab, **Report as (SWL)** — the identity that signs your reception reports and the PSK Reporter/WSPR uploads. It is separate from any callsign on purpose: it is never keyed, logged or spotted.

Part B — Listening

1. Pick a broadcast band on the band bar: **LW**, **MW**, **SW** or **FM** — or **AIR** and **MIL** for the civil and military airbands.
2. On shortwave the **metre band** is named — SW 49m · AM — and offered as a shortcut, so you can jump to a broadcast band's slice directly.
3. Pick the **mode**: **AM** for the broadcast bands, **SAM** for synchronous AM, **WFM** for FM broadcast.
4. **ECSS**. On **SAM**, the **ECSS-U** and **ECSS-L** presets keep one sideband and reject the other — the medium-wave DX trick for ducking an adjacent channel.
5. **Tone**. The SWL LOG window's **Tone** row shelves the bass, mid and treble of the demodulated audio, in front of the speakers. Broadcast audio wants a tone control the ham speech chain never needed.

6. **Missed something? REPLAY** plays the last two minutes instead of live — catch the station id you just heard.
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Part C — The broadcast schedule

The **SCHEDULE** window is the listener's worklist: it turns the same EiBi schedule that labels the waterfall into a table of about 4,600 broadcast transmissions. Filter it by a **UTC time** (or **now**), free text, **language**, **target** and a **metre band**, and star the ones you follow with ★ **FAVS**. A row can be **TUNEd** — the same tune as clicking its label on the waterfall — or **LOGged** straight into the reception log with the station, language and transmitter site already filled in. **SOLAR TIME** puts each row's local time at the *transmitter* beside its site.

Part D — The reception log (SWL LOG)

The **SWL LOG** window is your record of what was heard:

- **+ NEW** logs the station the receiver is on right now, with the live frequency, mode and S-meter reading; **LOG** on a SCHEDULE row fills in the rest. The entry form holds the station, frequency, mode, language, the **SINPO** or **SIO** figures, and your notes.
 - **Antenna** — the aerial in use, in your own words ("Longwire 20 m", "MLA-30 loop"). Set it once; each reception captures it.
 - **REPORT** writes the selected reception out as a text report to send to the broadcaster — the numbers, the **received-at** locator and the aerial, signed with your report identity and naming the program **sdroxide_SWL**.
 - **report sent** and **QSL received** track the whole loop, *hear* → *report* → *await QSL*: **REPORT** stamps the first, and a small **sent** or **QSL** mark on the row shows how far each catch has got.
 - **rcl** on a row tunes back to that reception's frequency and mode in one click, to check whether the station has returned.
 - **Pirate** marks an unlicensed broadcast; the row wears a **Jolly Roger**.
 - **Show** filters the list by free text, **band** and **day**, with a **Pirates only** switch. **CSV** and **ADIF** save the whole log — the ADIF as reception records, not contacts.
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Part E — Extras for listeners

- **SIG ID** names what is on the dial, ranked against the mode, frequency, band and passband from a built-in catalogue of ~60 signals, with a **sigidwiki** link for the sample.
- **JOBS** records a band on a timer — leave it running and come back to files named by UTC, frequency and mode.
- **Scanning** sweeps a band and names what it stops on.

- **Broadcast and utility labels** — EiBi transmitters, time signals, VOLMET and the rest — are drawn on the waterfall and on the panadapter.
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Help and links

- [README](#) — the full overview of the fork.
- [USER_MANUAL.md](#) — the manual, control by control.
- [Releases](#) — the latest build for every platform.

Good listening, and good DX! 73